SCIENCE / BC GRADE 11

Moles & molar mass

Convert between a substance's mass and amount.

Chemistry 11

01 The mole counts entities

One mole contains exactly $6.02214076 \times 10^{23}$ specified entities. These can be atoms, molecules, ions, or formula units. Always specify what is being counted.

02 Find molar mass

Add atomic molar masses according to the chemical formula. For H_2O, using H = 1.01 and O = 16.00 gives M = 18.02 g/mol. Use your class periodic table and its rounding conventions.

03 Use units to choose the operation

Amount n = mass m / molar mass M. Conversely, m = nM. In m/M, g divided by g/mol leaves mol. Keep extra digits during calculation and round the final result appropriately.

Worked example

Find the amount in 36.04 g of water

$M(H_2O) = 2(1.01) + 16.00 = 18.02$ g/mol. n = 36.04 g ÷ 18.02 g/mol = 2.000 mol of H_2O molecules.

Watch out: A mole is an amount, not a fixed mass. A mole of water and a mole of carbon dioxide contain the same number of molecules but have different masses.

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Try it. Then check it.

Work through these questions before looking at the answers.

1. Find the amount in 22.0 g CO_2 when M = 44.0 g/mol.

2. Find the mass of 0.250 mol NaCl when M = 58.44 g/mol.

3. About how many molecules are in 2.00 mol of water? Use 6.022×10^{23} per mole.

Answer key

1. n = 22.0/44.0 = 0.500 mol of CO_2.

2. m = 0.250 × 58.44 = 14.6 g to three significant figures.

3. $2.00 \times 6.022 \times 10^{23} = 1.20 \times 10^{24}$ molecules, to three significant figures.

One-minute recap

Close the notes. Explain one key idea in your own words, then retry the question you found most difficult.

About this guide

Original Study Bridge topic review prepared with AI assistance. Suggested course fit, not complete or official course coverage. Compare with your teacher's materials. Examples and practice questions are newly written.

Official course context: Chemistry 11

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